Tel Aviv University

The Iby and Aladar Fleischman Faculty of Engineering

FPGA Laboratory

Experiment FPGA 7 – Snake

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Based on Konstantin Berestizshevsky's work

Lab Goals (This is also the material for the test)

The purpose of this lab is to cover the following topics:

1. Design and implement logic from scratch, both combinatorial and synchronized, including block diagrams and module instantiations.
2. Utilize previous labs code regarding VGA, PS/2, 7-segment display, counter etc.

Submission Guidelines

The lab consists of TBD tasks, where each task depends on the previous.

1. Submit a report (PDF format), as well as the XDC constraints file, and all the *.v files you have edited. All zipped together in a zip file named FPGA7 <ID1> <ID2>
2. Each waveform must be annotated by (i) drawing arrows that point to important signals and (ii) explaining these signals.
3. Most of the modules in this lab are parametric (for example, Compadder(N) has a design parameter N, which determines the bit-width of its operands). These modules must in fact support different parameter values, and not to stick to some constant value.
4. Each Verilog source/test-bench header must have both of your names in it:

```
`timescale 1ns/10ps
//////////////////////////////////////////////
// Company:          Tel Aviv University
// Engineer:         Your name
```

5. At home you can work with the [Vivado Web-Pack](). You will need to create a free student account in Xilinx before downloading and installing the software.
6. It is your responsibility to either use a USB-flash drive or other cloud solutions to save your project before you leave the laboratory. The laboratory PCs are erased from time to time.

1. Vivado project

Launch the Vivado software, create a new project. Follow the "Vivado Project Creation" subsection of the Vivado-tutorial to set up a new project. The name of the project should be **lab7**, and it should be saved in **G:\FPGA_LAB** path. The project type is **RTL Project**. Make sure to choose the **xc7a35tlcpg236-2L** part in the "Default Part" screen of the wizard. Notes:

- To add more sources later, go to "Project Manager→Add Sources" in the flow-navigator in the left part of the Vivado screen.
- In this lab, all the tasks are to be performed under the same "lab7" project.

Snake game implementation

Design and implement a snake game using the Basys3 FPGA board. The game will display graphics on a computer monitor via a VGA interface, count and display the score on the 7-segment display, and allow player control through the keyboard.

Requirements:

1) **Game Display**
   a. Use the VGA interface to visualize the snake game on a computer monitor
   b. The snake and food should be clearly distinguishable
   c. The screen should refresh at a sufficient rate to ensure smooth gameplay
2) **Score Display**
   a. Use the 7-segment display to show the current score
   b. The score should increment when the snake eats the food
3) **Control**
   a. Use the keyboard buttons to control the snake's direction
      i. 4 = Left
      ii. 6 = Right
      iii. 8 = Up
      iv. 5 = Down
   b. Ensure there is no immediate reversal of the snake's direction (e.g., from left to right directly)
4) **Game Logic**
   a. Implement game over conditions (e.g., snake runs into itself or the wall)
   b. **Bonus** - Create a game over screen to display after game over conditions

Guidance: In this task, you are expected to design and implement the modules by yourselves without detailed guidance. We will not supply any code or module inputs and outputs. This exercise is meant to encourage independent problem-solving and application of the concepts learned in previous labs. However, we provide the following high-level guidance to help you get started:

1) **Understand the Requirements**
   a. Carefully read through the task requirements and ensure you understand what each component of the game needs to do
2) **Design a Plan**
   b. Break down the task into smaller modules, such as the VGA interface, snake logic, food logic, collision detection, score display and more.
   c. Create a high-level block diagram of your design showing how these modules will interact
3) **Leverage Previous Labs**
   d. Review the VGA interface from the previous lab. You will need to modify and extend it to display the snake and food
   e. Utilize the knowledge gained from working with the 7-segment display for the score counter

f. Utilize the keyboard interface to manage the user controllability

4) **Design Each Module**
    g. For each module, define its inputs, outputs, and functionality
    h. Start with simple versions of each module and iteratively add complexity
5) **Test Incrementally**
    i. Test each module individually before integrating them into the complete system
    j. Use simulation tools to verify the functionality of your modules

Report Submission Guidance

6) **Design Overview**
    a. High-level block diagram
    b. Description of the overall design and how the modules interact
7) **Module Descriptions**
    a. Detailed explanation of each module, including:
        i. Purpose and functionality
        ii. Inputs and outputs
        iii. Internal design and logic
    b. Discuss how each module was tested individually
8) **Utilization of Previous Labs**
    a. Explanation of how the concepts and modules from previous labs were utilized in this project
    b. Specific examples of code or design elements reused or adapted
9) **Results**
    a. Summary of the final working state of the game
    b. Screenshots or photos showing the game in action

In the Lab:

Connect the Jtag cable from PC to FPGA board
Connect the VGA cable from the monitor to the FPGA board.
Program the FPGA with your implemented design and call an instructor to show him a correct functioning of your game.
Take a photo of a correct functioning of the monitor.

Submit:

In the report – submit all the Verilog files:
Provide a screenshot of the simulation and explain its principle signals behavior by pointing with an arrow to these signals' transitions and annotating them.
Explain which problems appeared during the lab and how did you overcome these problems. Attach a photo of the correct monitor operation.